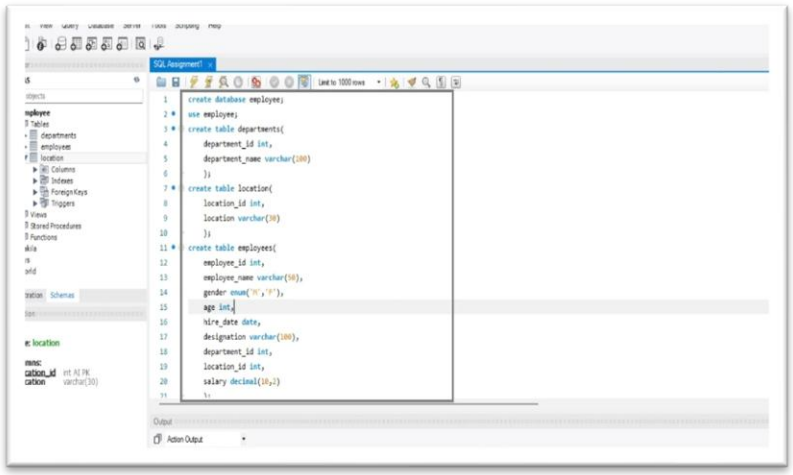
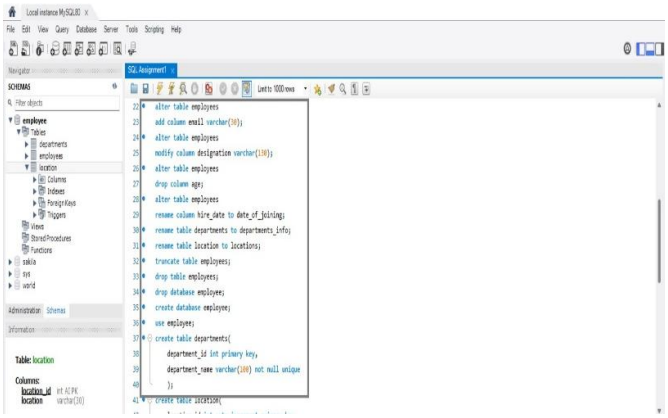
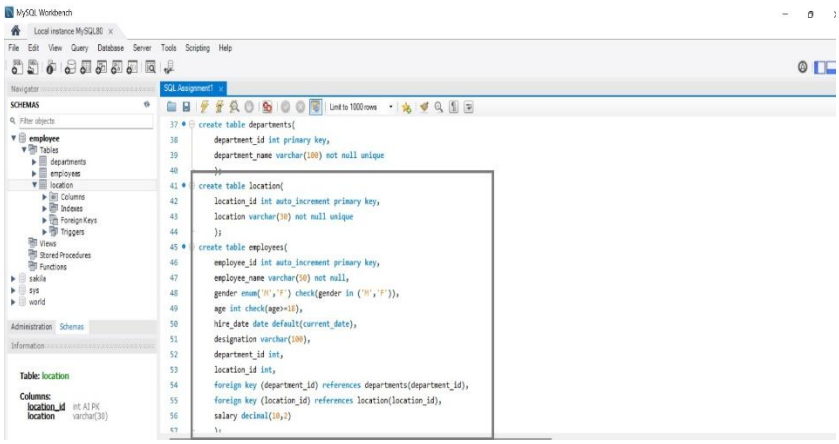
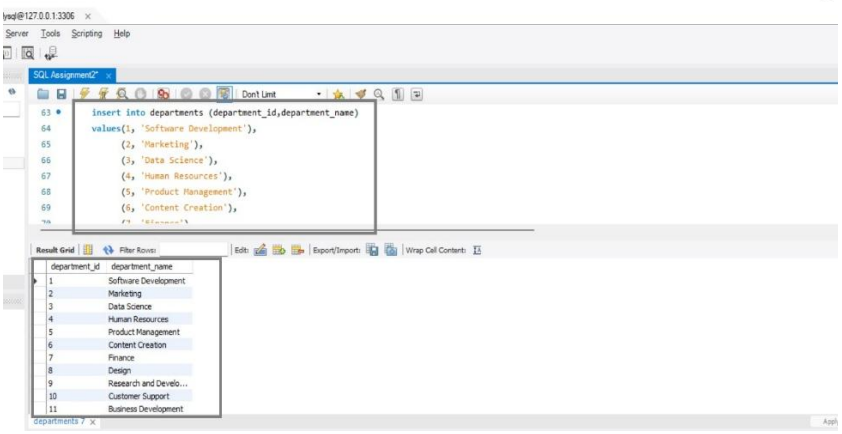
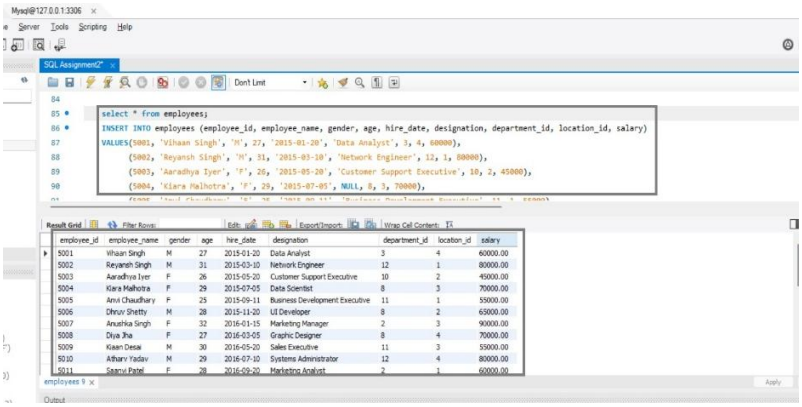
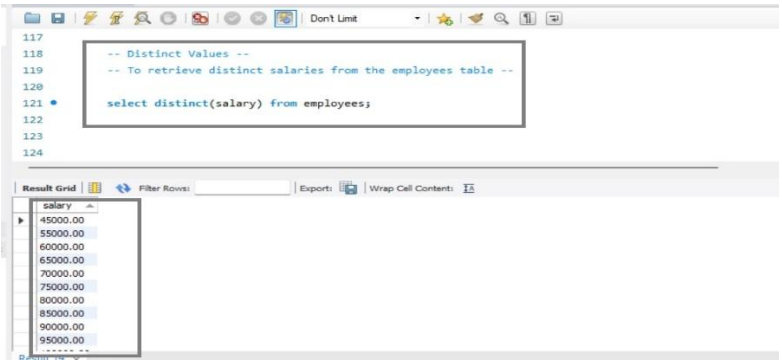

SQL Assignment1
create database employee;
use employee;
create table departments(
department_id int,
department_name varchar(100)
);
create table location(
location_id int,
location varchar(30)
);
create table employees(
employee_id int,
employee_name varchar(50),
gender enum('M','F'),
age int,
hire_date date,
designation varchar(100),
department_id int,
location_id int,
salary decimal(10,2)
);
Output
Action Output

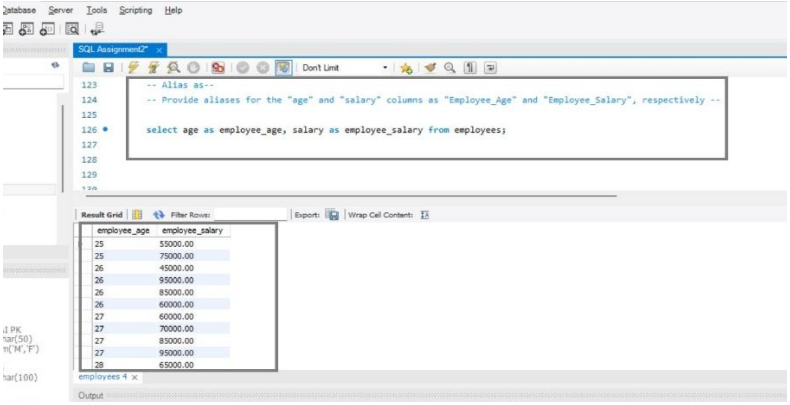

Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL Assignment1
SCHEMAS
employee
Tables
departments
employees
location
Columns
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Table: location
Columns:
location_id int AI PK
location varchar(30)
alter table employees
add column email varchar(30);
alter table employees
modify column designation varchar(130);
alter table employees
drop column age;
alter table employees
rename column hire_date to date_of_joining;
rename table departments to departments_info;
rename table location to locations;
truncate table employees;
drop table employees;
drop database employee;
create database employee;
use employee;
create table departments(
department_id int primary key,
department_name varchar(100) not null unique
);

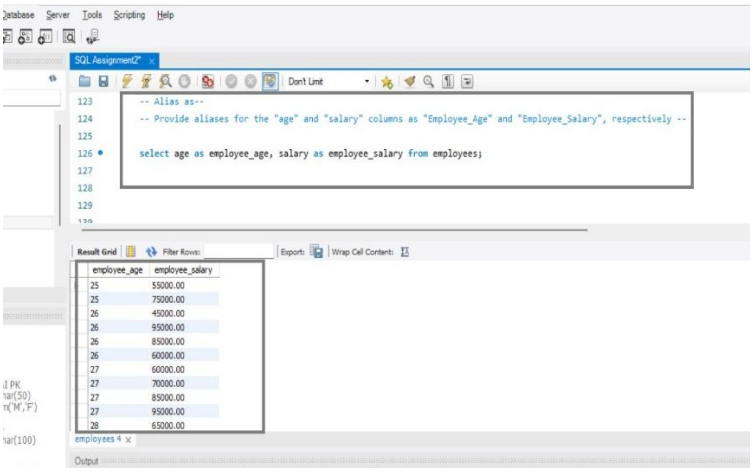

MySQL Workbench
Local instance MySQL80
File Edit View Query Database Server Tools Scripting Help
SQL Assignment1
SCHEMAS
employee
Tables
departments
employees
location
Columns
Indexes
Foreign Keys
Triggers
Views
Stored Procedures
Functions
sakila
sys
world
Administration Schemas
Information
Table: location
Columns:
location_id int AI PK
location varchar(30)
create table departments(
department_id int primary key,
department_name varchar(100) not null unique
);
create table location(
location_id int auto_increment primary key,
location varchar(30) not null unique
);
create table employees(
employee_id int auto_increment primary key,
employee_name varchar(50) not null,
gender enum('M','F') check(gender in ('M','F')),
age int check(age>18),
hire_date date default(current_date),
designation varchar(100),
department_id int,
location_id int,
foreign key (department_id) references departments(department_id),
foreign key (location_id) references location(location_id),
salary decimal(10,2)
);

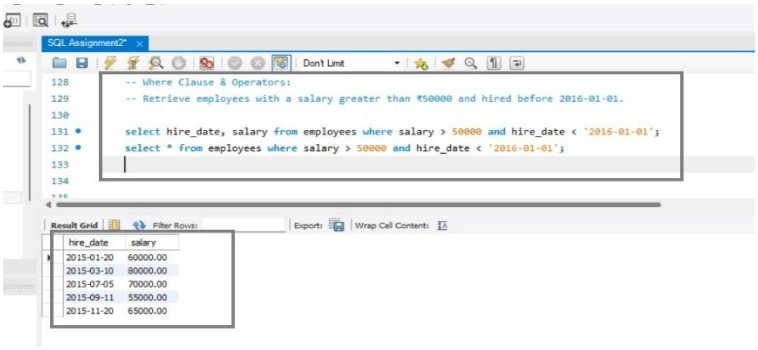

SQL Assignment2*
insert into departments (department_id,department_name)
values(1, 'Software Development'),
(2, 'Marketing'),
(3, 'Data Science'),
(4, 'Human Resources'),
(5, 'Product Management'),
(6, 'Content Creation'),
department_id department_name
1 Software Development
2 Marketing
3 Data Science
4 Human Resources
5 Product Management
6 Content Creation
7 Finance
8 Design
9 Research and Develo...
10 Customer Support
11 Business Development

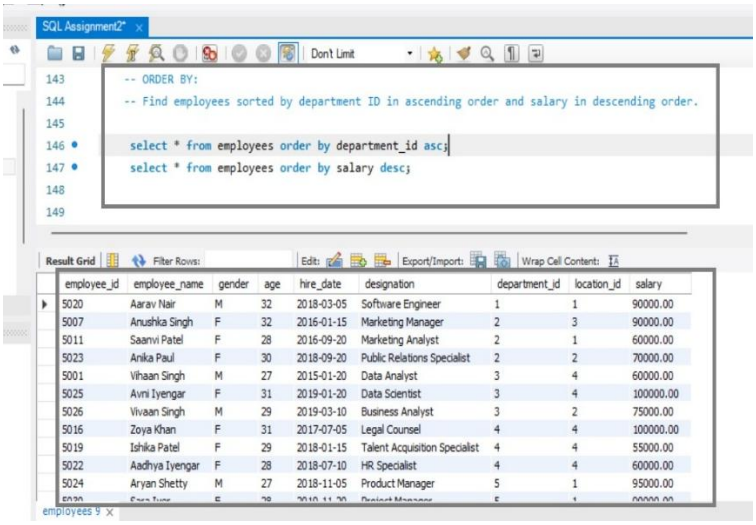

SQL Assignment2*
select * from employees;
INSERT INTO employees (employee_id, employee_name, gender, age, hire_date, designation, department_id, location_id, salary)
VALUES(5001, 'Vihaan Singh', 'M', 27, '2015-01-20', 'Data Analyst', 3, 4, 60000),
(5002, 'Reyansh Singh', 'M', 31, '2015-03-10', 'Network Engineer', 12, 1, 80000),
(5003, 'Aaradhya Iyer', 'F', 26, '2015-05-20', 'Customer Support Executive', 10, 2, 45000),
(5004, 'Kiara Malhotra', 'F', 29, '2015-07-05', NULL, 8, 3, 70000),
employee_id employee_name gender age hire_date designation department_id location_id salary
5001 Vihaan Singh M 27 2015-01-20 Data Analyst 3 4 60000.00
5002 Reyansh Singh M 31 2015-03-10 Network Engineer 12 1 80000.00
5003 Aaradhya Iyer F 26 2015-05-20 Customer Support Executive 10 2 45000.00
5004 Kiara Malhotra F 29 2015-07-05 Data Scientist 8 3 70000.00
5005 Anvi Chaudhary F 25 2015-09-11 Business Development Executive 11 1 55000.00
5006 Dhruv Shetty M 28 2015-11-20 UI Developer 8 2 65000.00
5007 Anushka Singh F 32 2016-01-15 Marketing Manager 2 3 90000.00
5008 Diya Jha F 27 2016-03-05 Graphic Designer 8 4 70000.00
5009 Kiaan Desai M 30 2016-05-20 Sales Executive 11 3 55000.00
5010 Atharv Yadav M 29 2016-07-10 Systems Administrator 12 4 80000.00
5011 Saanvi Patel F 28 2016-09-20 Marketing Analyst 2 1 60000.00

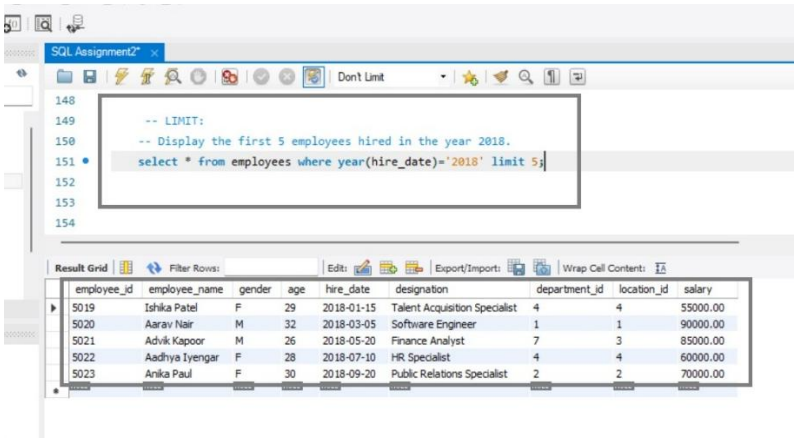

-- Distinct Values --
-- To retrieve distinct salaries from the employees table --
select distinct(salary) from employees;
salary
45000.00
55000.00
60000.00
65000.00
70000.00
75000.00
80000.00
85000.00
90000.00
95000.00

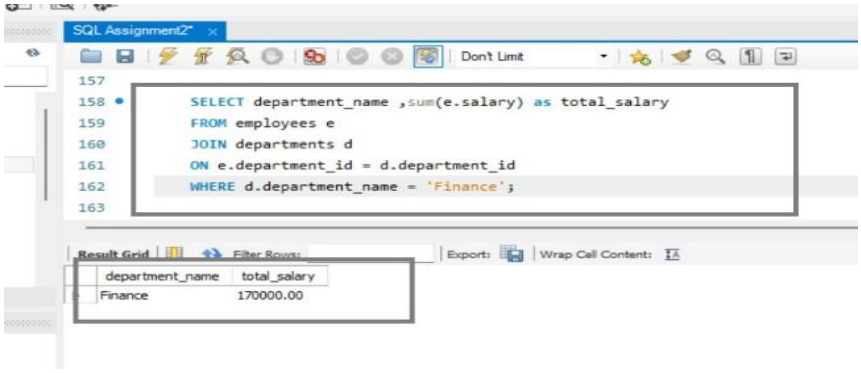

SQL Assignment2*
-- Alias as--
-- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively --
select age as employee_age, salary as employee_salary from employees;
employee_age employee_salary
25 55000.00
25 75000.00
26 45000.00
26 95000.00
26 85000.00
26 60000.00
27 60000.00
27 70000.00
27 85000.00
27 95000.00
28 65000.00

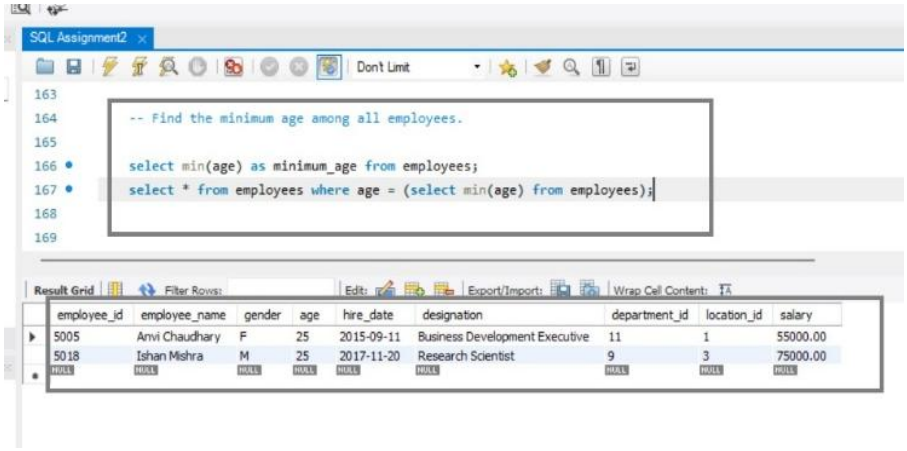

Database Server Tools Scripting Help
SQL Assignment2*
Don't Limit
-- Alias as--
-- Provide aliases for the "age" and "salary" columns as "Employee_Age" and "Employee_Salary", respectively --
select age as employee_age, salary as employee_salary from employees;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
employee_age employee_salary
25 55000.00
25 75000.00
26 45000.00
26 95000.00
26 85000.00
26 60000.00
27 60000.00
27 70000.00
27 85000.00
27 95000.00
28 65000.00
employees 4
Output

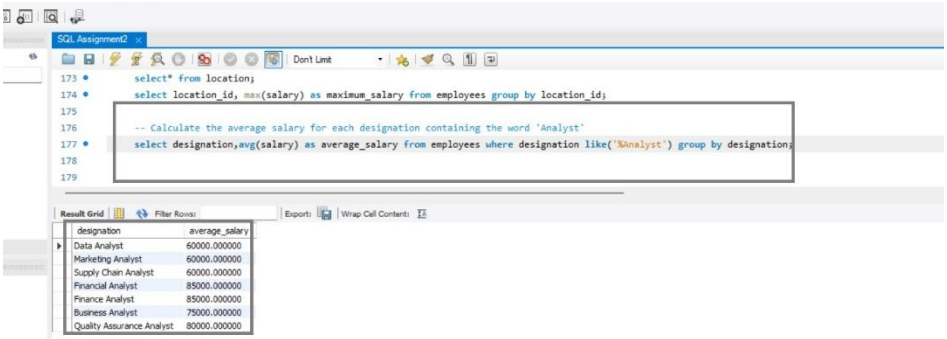

SQL Assignment2*
Don't Limit
-- Where Clause & Operators:
-- Retrieve employees with a salary greater than ₹50000 and hired before 2016-01-01.
select hire_date, salary from employees where salary > 50000 and hire_date < '2016-01-01';
select * from employees where salary > 50000 and hire_date < '2016-01-01';
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
hire_date salary
2015-01-20 60000.00
2015-03-10 80000.00
2015-07-05 70000.00
2015-09-11 55000.00
2015-11-20 65000.00


SQL Assignment2*
Don't Limit
-- ORDER BY:
-- Find employees sorted by department ID in ascending order and salary in descending order.
select * from employees order by department_id asc;
select * from employees order by salary desc;
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
5020 Aarav Nair M 32 2018-03-05 Software Engineer 1 1 90000.00
5007 Anushka Singh F 32 2016-01-15 Marketing Manager 2 3 90000.00
5011 Saanvi Patel F 28 2016-09-20 Marketing Analyst 2 1 60000.00
5023 Anika Paul F 30 2018-09-20 Public Relations Specialist 2 2 70000.00
5001 Vihaan Singh M 27 2015-01-20 Data Analyst 3 4 60000.00
5025 Avni Iyengar F 31 2019-01-20 Data Scientist 3 4 100000.00
5026 Vivaan Singh M 29 2019-03-10 Business Analyst 3 2 75000.00
5016 Zoya Khan F 31 2017-07-05 Legal Counsel 4 4 100000.00
5019 Ishika Patel F 29 2018-01-15 Talent Acquisition Specialist 4 4 55000.00
5022 Aadhya Iyengar F 28 2018-07-10 HR Specialist 4 4 60000.00
5024 Aryan Shetty M 27 2018-11-05 Product Manager 5 1 95000.00
employees 9


SQL Assignment2*
Don't Limit
148
149 -- LIMIT:
150 -- Display the first 5 employees hired in the year 2018.
151 • select * from employees where year(hire_date)='2018' limit 5;
152
153
154
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
5019 Ishika Patel F 29 2018-01-15 Talent Acquisition Specialist 4 4 55000.00
5020 Aarav Nair M 32 2018-03-05 Software Engineer 1 1 90000.00
5021 Advik Kapoor M 26 2018-05-20 Finance Analyst 7 3 85000.00
5022 Aadhya Iyengar F 28 2018-07-10 HR Specialist 4 4 60000.00
5023 Anika Paul F 30 2018-09-20 Public Relations Specialist 2 2 70000.00


SQL Assignment2*
Don't Limit
157
158 • SELECT department_name ,sum(e.salary) as total_salary
159 FROM employees e
160 JOIN departments d
161 ON e.department_id = d.department_id
162 WHERE d.department_name = 'Finance';
163
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
department_name total_salary
Finance 170000.00

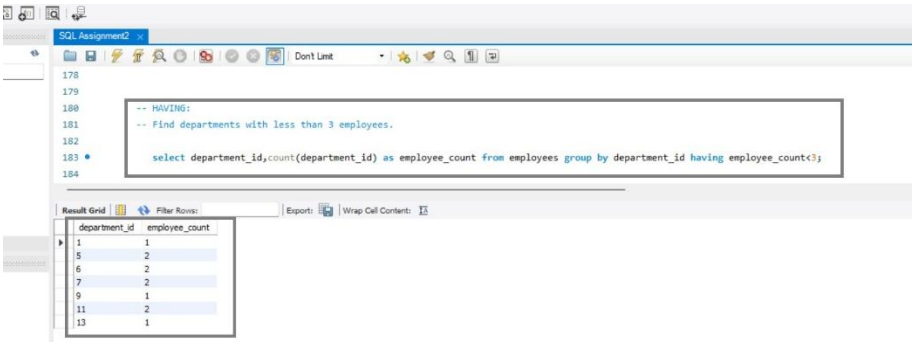

SQL Assignment2
Don't Limit
163
164 -- Find the minimum age among all employees.
165
166 • select min(age) as minimum_age from employees;
167 • select * from employees where age = (select min(age) from employees);
168
169
Result Grid
Filter Rows:
Edit:
Export/Import:
Wrap Cell Content:
employee_id employee_name gender age hire_date designation department_id location_id salary
5005 Anvi Chaudhary F 25 2015-09-11 Business Development Executive 11 1 55000.00
5018 Ishan Mishra M 25 2017-11-20 Research Scientist 9 3 75000.00



```
select* from location;
select location_id, max(salary) as maximum_salary from employees group by location_id;

-- Calculate the average salary for each designation containing the word 'Analyst'
select designation,avg(salary) as average_salary from employees where designation like('%Analyst') group by designation;
```

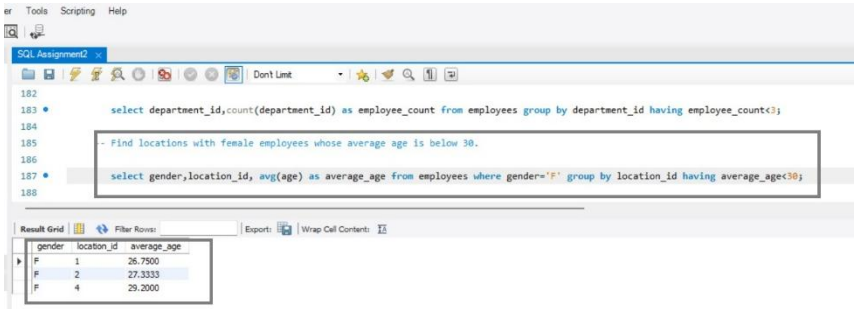
designation	average_salary
Data Analyst	60000.000000
Marketing Analyst	60000.000000
Supply Chain Analyst	60000.000000
Financial Analyst	85000.000000
Finance Analyst	85000.000000
Business Analyst	75000.000000
Quality Assurance Analyst	80000.000000



```
-- HAVING:
-- Find departments with less than 3 employees.

  select department_id,count(department_id) as employee_count from employees group by department_id having employee_count<3;
```

department_id	employee_count
1	1
5	2
6	2
7	2
9	1
11	2
13	1

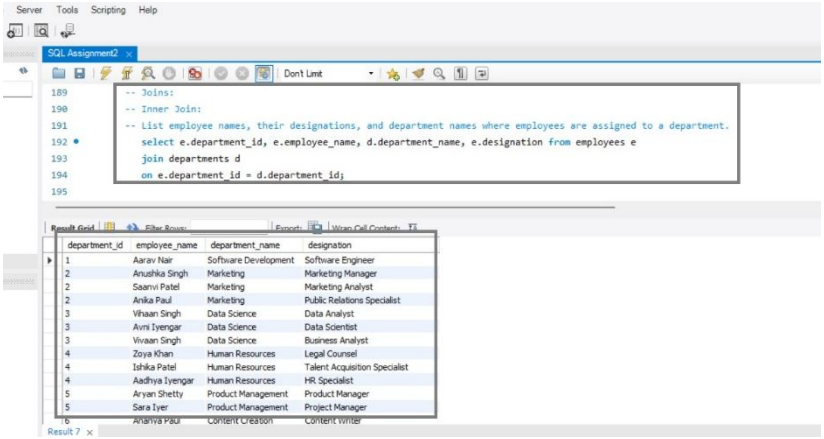


```
select department_id,count(department_id) as employee_count from employees group by department_id having employee_count<3;

- Find locations with female employees whose average age is below 30.

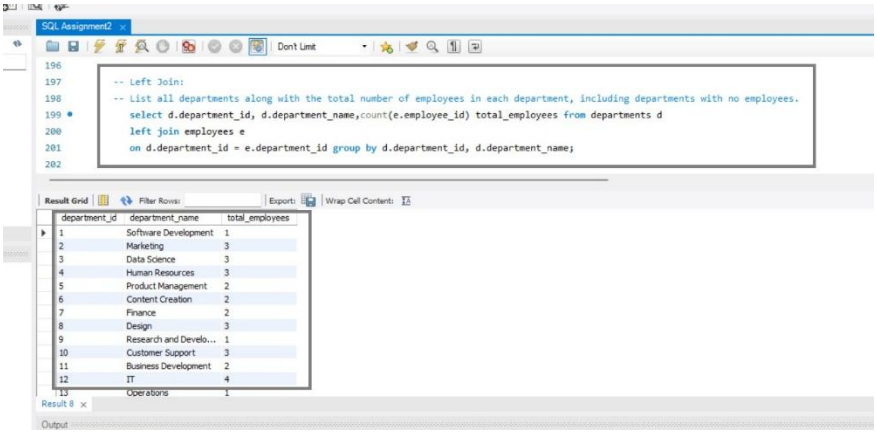
select gender,location_id, avg(age) as average_age from employees where gender='F' group by location_id having average_age<30;
```

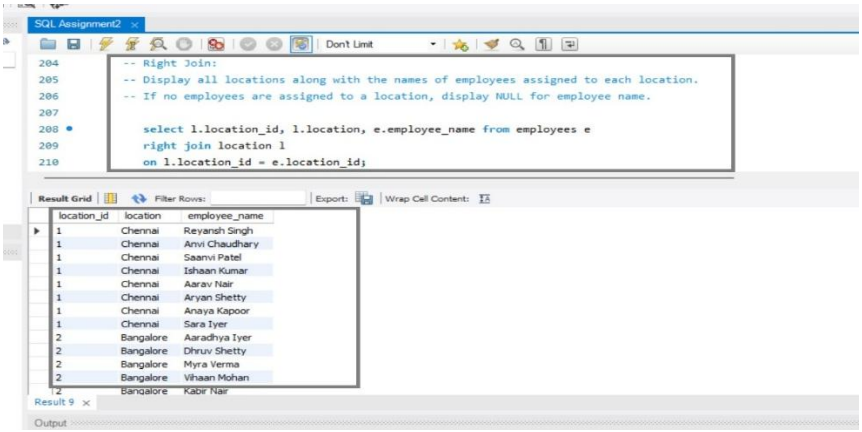
gender	location_id	average_age
F	1	26.7500
F	2	27.3333
F	4	29.2000



```
-- Joins:
-- Inner Join:
-- List employee names, their designations, and department names where employees are assigned to a department.
   select e.department_id, e.employee_name, d.department_name, e.designation from employees e
   join departments d
   on e.department_id = d.department_id;
```

department_id	employee_name	department_name	designation
1	Aarav Nair	Software Development	Software Engineer
2	Anushka Singh	Marketing	Marketing Manager
2	Saanvi Patel	Marketing	Marketing Analyst
2	Anika Paul	Marketing	Public Relations Specialist
3	Vihaan Singh	Data Science	Data Analyst
3	Avni Iyengar	Data Science	Data Scientist
3	Vivaan Singh	Data Science	Business Analyst
4	Zoya Khan	Human Resources	Legal Counsel
4	Ishika Patel	Human Resources	Talent Acquisition Specialist
4	Aadhya Iyengar	Human Resources	HR Specialist
5	Aryan Shetty	Product Management	Product Manager
5	Sara Iyer	Product Management	Project Manager
6	Ananya Paul	Content Creation	Content Writer


SQL Assignment2
Don't Limit
-- Left Join:
-- List all departments along with the total number of employees in each department, including departments with no employees.
select d.department_id, d.department_name,count(e.employee_id) total_employees from departments d
left join employees e
on d.department_id = e.department_id group by d.department_id, d.department_name;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
department_id | department_name | total_employees
1 | Software Development | 1
2 | Marketing | 3
3 | Data Science | 3
4 | Human Resources | 3
5 | Product Management | 2
6 | Content Creation | 2
7 | Finance | 2
8 | Design | 3
9 | Research and Develo... | 1
10 | Customer Support | 3
11 | Business Development | 2
12 | IT | 4
13 | Operations | 1
Result 8
Output


SQL Assignment2
Don't Limit
-- Right Join:
-- Display all locations along with the names of employees assigned to each location.
-- If no employees are assigned to a location, display NULL for employee name.
select l.location_id, l.location, e.employee_name from employees e
right join location l
on l.location_id = e.location_id;
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
location_id | location | employee_name
1 | Chennai | Reyansh Singh
1 | Chennai | Anvi Chaudhary
1 | Chennai | Saanvi Patel
1 | Chennai | Ishaan Kumar
1 | Chennai | Aarav Nair
1 | Chennai | Aryan Shetty
1 | Chennai | Anaya Kapoor
1 | Chennai | Sara Iyer
2 | Bangalore | Aaradhya Iyer
2 | Bangalore | Dhruv Shetty
2 | Bangalore | Myra Verma
2 | Bangalore | Vihaan Mohan
2 | Bangalore | Kabir Nair
Result 9
Output